

CY 304: Fundamental Analytical Chemistry

Course Code	CY 304
Course Name	Fundamental Analytical Chemistry
L-T-P-C	3-0-0-3
Prerequisites	None
Intended for	BS Chemical Sciences
Mutual Exclusion	None
Approval	42nd Senate (Jan 2024)

Course Contents

- **Basic Tools of Analytical Chemistry:** Measurements in Analytical Chemistry, Units of Measurement, Uncertainty in Measurements, Concentration, Molarity and Formality, Normality, Molality, Weight, Volume, and Weight-to-Volume Percent, Parts Per Million and Parts Per Billion, Converting Between Concentration Units, Stoichiometric Calculations, Types of errors in Chemical Analyses, Accuracy, Precision, Sensitivity, Specificity and Selectivity, Sampling, Standardization and Calibration, Least square fit, Limit of detection and quantification, Statistical Data Treatment and Evaluation, Basic Equipment, Signal and noise in instrumental measurement, Equipment for Measuring Mass and Volume, Equipment for Drying Samples, Spreadsheets and Computational Software . (10 Hours)
- **Classical Methods of Analysis:** Preparing Solutions, Preparing Stock Solutions, Preparing Solutions by Dilution, Gravimetric Methods of Analysis, Titrations in Analytical Chemistry, Principles of Neutralization Titrations, Complex Acid/Base Systems, Applications of Neutralization Titrations, Complexation and Precipitation Reactions and Titrations, Titration Curves, Calculation of the pH of Solutions. (10 Hours)
- **Chemical Equilibria:** Reversible Reactions and Chemical Equilibria, Thermodynamics and Equilibrium Chemistry, Equilibrium Constants for Chemical Reactions, Precipitation Reactions, Acid– Base Reactions, Complexation Reactions, Oxidation–Reduction (Redox) Reactions, Le Châtelier’s Principle, Buffer Solutions. (8 Hours)
- **Analytical Electrochemistry:** Potentiometry-General principles, Calomel Electrodes, Ag-AgCl electrodes, Membrane electrodes-ion selective electrodes, glass electrodes, biosensors. Coulometry: Basic principles, constant current and constant potential coulometry. Voltammetry: different waveforms – linear scan, square scan and triangular scan, cyclic voltammetry. (8 Hours)
- **Separation Techniques:** Principles and applications of TLC, General Theory of Column Chromatography, Gas chromatography (GC), High Performance Liquid Chromatography (HPLC), FPLC, Ion chromatography (IC), Supercritical Fluid Chromatography, Capillary Electrophoresis. (6 Hours)

Text Books

1. D. A. Skoog, D. M. West, F. J. Holler, S. R. Crouch, **Fundamentals of Analytical Chemistry**, 9th Edition, Thomson, 2013.
2. D. Harvey, **Analytical chemistry 2.1**, McGraw-Hill, 2016.
3. G.D. Christian, P.K. Dasgupta and K.A. Schug, **Analytical Chemistry**, 7th edition, Wiley
4. H. H. Willard, L. L. Merritt Jr., J. A. Dean, f. A. Settle Jr., **Instrumental Methods of Analysis**, CBS Publishers, 1986.
5. J. C. Miller, J. N. Miller, **Statistics for Analytical Chemistry**, 2nd Edition, Wiley, 1998.
6. D. C. Harris, **Quantitative Chemical Analysis**, 7th Edition, W. H. Freeman, 2006.

References

1. NA